

Amogh Parab

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SUMMARY

Mathematics PhD from *the* Ohio State University with more than three years of experience in both object-oriented programming (Python) and functional programming (Lean4) • **Contributed to open-source projects** and currently formalizing mathematical research problems to **support AI training pipelines at Harmonic** • Passionate about formal methods and their applications to scientific discovery and AI safety.

SKILLS

- Formal Methods: Lean4 (Mathlib, Verified-zkEVM-CompPoly), dependent type theory, formal verification, theorem proving
- Languages: Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), LaTeX, Julia
- Mathematics: Category Theory, Algebraic Topology, Abstract Algebra, Type Theory, Numerical Methods
- OS and Platforms: Bash, GitHub, Windows, Linux, Jupyter Notebook, Visual Studio Code

WORK EXPERIENCE

Harmonic

Remote

Lean Expert (part-time)

Sep. 2025 – Present

- **Formalized over 200 advanced mathematics problems** in Lean4, spanning a broad range of topics, including algebra, geometry, and number theory to support the development of an advanced mathematical reasoning model.
- **Contributed to open-source formalization projects**, including Mathlib, Google DeepMind's Formal Conjectures, and Verified-zkEVM-CompPoly, expanding the frontier of machine-verifiable mathematics.
- Designed and implemented a Euclidean Geometry library in Lean4 to streamline formalization workflows.

The Ohio State University

Columbus, Ohio

Graduate Teaching Associate / Lecturer

Aug. 2020 – May 2026

- Created and presented over **150 hours of mathematics content**, incorporating visual demonstrations to enhance student engagement, with an average teaching score of 4.7/5.
- Led Engineering Mathematics recitations, conducted office hours, proctored exams, and prepared teaching materials for undergraduate engineering students.

PROJECT EXPERIENCE

Rewriting Systems in Lean4 ([GitHub](#))

Columbus, Ohio

The Ohio State University

Aug. 2023 – Jan. 2025

- **Designed an inductive data type in Lean4** to serve as a rewriting system for multiplicative words, a core component of the categorical group structures in my PhD thesis. Extended Mac Lane's coherence theorem to categorical groups.
- Formally verified fundamental algebraic properties, including **associative concatenation and the universal property**; authored over **2,000 lines of Lean4 code** and **20+ pages of technical documentation in LaTeX**.

Formalizing Homotopy Lifting Property ([GitHub](#))

Berkeley, California

SLMath (formerly MSRI)

Jun. 2023

- **Collaborated with Mathlib maintainers** at SLMath to port the Lean mathematics library from Lean3 to Lean4 during a two-week formalization summer school.
- **Formalized the uniqueness of the homotopy lifting property** in Lean4, writing over 300 lines of code; presented findings alongside a team of six at the closing session.

Data Science Bootcamp (Project 1: [GitHub](#)) (Project 2: [GitHub](#))

Remote

Participant at the Erdős Institute

Sep. 2024 – Nov. 2024

- Predicted results of Valorant online matches with **67% accuracy using XGBClassifier** on selected key features from 160 engineered features; applied **grid search for hyperparameter tuning**.
- Developed a predictive model for car prices with an R^2 value of 0.88 using XGBoost in a team of four, **achieving a 22% improvement** over the baseline model; utilized SHAP values to identify key feature importance.

Creep Evolution in FCC crystalline structure <u>Texas A&M University (Internship)</u> Analyzed stress-strain relationship of exposed creep on external boundary conditions using ABAQUS CAD software.	College Station, Texas May 2017 – Jul. 2017
Quasi-Solution Approach to Non-Linear Differential Equations <u>The Ohio State University (Internship)</u> Provided analytic solution to the Blasius boundary condition (Fluid Dynamics) using Maple software. Used the Banach fixed-point theorem to show the existence of the solution.	Columbus, Ohio Aug. 2016 – Jan. 2017
Optimal Stochastic Growth Model Using U.S. Data (Penn World Table) (GitHub) <u>Personal Project</u> Implemented a discrete-time stochastic optimal growth model in Julia using real macroeconomic data from the Penn World Table (PWT). Compared the theoretical optimal policy to observed outcomes.	Columbus, Ohio Nov. 2025 – Dec. 2025

PUBLICATIONS

- “Coherence and Symmetrization of Categorical Groups”, PhD thesis, The Ohio State University, 2025
- “Gröbner bases of Rational Normal Curves”, Proceedings of the COCOA Workshop, IIT Gandhinagar, 2019

GRANTS AND AWARDS

- Special Graduate Assignment for Thesis Research, The Ohio State University, 2023
- Bronze Medal for the best performance in the core courses of Mathematics, IIT Gandhinagar, 2019
- Dean’s List, IIT Gandhinagar, Spring 2015, Fall 2015, Spring 2016

EDUCATION

The Ohio State University PhD in Mathematics (Cumulative GPA: 3.99/4) Thesis: Coherence and Symmetrization of Categorical Groups Coursework: Mathematics of Data Science, Abstract Algebra, Analysis, Lie Theory, Topological Quantum Field Theory	Columbus, Ohio Aug. 2025
Indian Institute of Technology Gandhinagar BTech in Mechanical Engineering, MSc in Mathematics (Cumulative GPA: 3.90/4) Master’s Thesis: On Riemann-Roch Theorem Coursework (Master’s): Linear Algebra, Numerical Methods, Ordinary and Partial Differential Equations, Group Theory Coursework (BTech): Quantum Mechanics, Non-Linear Dynamics	Gandhinagar, India Jul. 2019

LEADERSHIP EXPERIENCE

Directed Reading Program <u>Program Coordinator, The Ohio State University</u> - Organized the pairing process of more than 100 undergraduate mathematics enthusiasts with experienced graduate mentors to explore topics of mutual interest. Mentored five students over two years in advanced mathematical concepts beyond the standard curriculum; introduced formal theorem proving via Lean’s Natural Number Game.	Columbus, Ohio Aug. 2021 – May 2025
Team LaTeX <u>Co-founder, IIT Gandhinagar</u> Co-founded a LaTeX club at IIT Gandhinagar to train students and researchers in professional scientific typesetting; organized week-long workshops attended by hundreds of students and faculty.	Gandhinagar, India Aug. 2017 – May 2018